

Exploring Climate Risk Analogues in North America

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Abstract

While climate analogues are widely used in climate adaptation, the incorporation of risk measurements into the analogue matching process has not been widely explored. Studying the probability of flood occurrences as a climate risk, this study builds upon the concept of analogues by predicting present and future flood risks for the year 2050 under the climate change scenarios of RCP 4.5 and RCP 8.5. In addition to random forests, linear models such as binary and ordinal logistic regression, as well as Poisson regression, are trained to predict 1) the probability of a flood, 2) the probability of a category 1, 2, or 3 flood, 3) the death count of a flood, 4) the displacement count of a flood. The analogues are calculated within the boundaries of the North American region, but grids are of low resolution at 0.22° latitude and longitude. Along with errors accumulated from the interpolation of the predictors of temperature, precipitation and elevation, training processes of the above machine learning models are heavily flawed, thus impacting the resulting flood risk predictions. Subsequently, many analogue calculations led to present-day location matches that are unrealistic, but with more improvements in interpolation and model training, flood risk analogues can provide new insights into climate risk mitigation.

Contents

Section I: Introduction and Literature Review	3
Section II: Methodology	4
Section 2.1: Definition of Flood Risk	4
Section 2.2: Geospatial Data Organization	4
Section 2.3: Building 2025 Rasters	5
Section 2.4: Building 2050 Rasters	7
Section 2.5: Model Training	8
Section 2.6: Prediction of Flood Risks	11
Section 2.7: Calculating Analogue Locations	13
Section 2.8: Model Assumptions	14
Section III: Results	15
Section 3.1: Model Validation and Predictive Performance	15
Section 3.2: Analogue Index	16
Section IV: Discussion and Conclusion	18
Section 4.1: Applications of Flood Risk Analogues	18
Section 4.2: Potential Improvements	18
References	20

Section I: Introduction and Literature Review

Climate analogues, in terms of temperature, have been widely studied due to their applications in urban planning, policymaking and stormwater engineering (Fitzpatrick and Dunn, 2019). By finding a representative city that, at present time, reflects the future climate of a target city, researchers can reference the urban planning designs of analogue cities that are already enduring the impacts of climate change in the target city's adaptation plans. Often presented on the global scale, climate analogue maps can provide matches of locations that share similar topography, climate, and ecology despite being separated by continents (Fitzpatrick, 2019). Under various policy scenarios, such as between Representative Concentration Pathways (RCP) 1 and 4, the target city's respective analogue will also be vastly different due to the impact of radiative forcing on the sensitivity of temperature change in each location.

However, just referencing an analogue city that shares a similar climate does not give a comprehensive insight into the potential risks brought on by climate change. Due to data scarcity for historic observed temperature, precipitation and flood events on the global scale, and limitations on resolution imposed by the utilization of a global scale, climate analogues are often constrained to just one measurement (often temperature) in the assessment of similarities between locations. As of 2025, a thorough compilation of global climate analogues involving a risk measurement on flooding, wildfire, and mortality has yet to exist.

Expanding upon the concept of global climate analogues, this study explores analogue locations and their potential applications in risk mitigation by focusing on floods as an example of climate risk. Floods are still considered rare, extreme events in that their risks are hard to quantify for both the present time and the future (Wu et al., 2024). The study of flood risks, also known as the probability of flood occurrence, have relied mainly on the usage of supervised machine learning models, such as logistic regression and random forests, in future risk prediction. Model training and predictions are often constrained to the scale of a city or a province, global scale studies of flood risks have rarely been shown to be compatible with linear models (Kundzewicz et al., 2013).

The year 2050 is often defined as a key year for climate policy. The Intergovernmental Panel on Climate Change identifies this year as a target in which many developed countries must achieve net zero emissions (IPCC, 2018) for average global temperature increase to stay within 2°C. If this target year is not met, then RCP scenarios of 1 to 2 are highly unlikely. Given the changes in flood risks brought on by climate change, this study will specifically examine present-day analogue locations for the flood risks of the key year of 2050.

This study will focus solely on the continent of North America due to the absence of a comprehensive database of historic temperature, precipitation and flood event observations on the global scale. As linear models have been shown to have better predictive performance with provincial boundaries, the author of this study posits that linear models will also be successful on a regional scale (Wahba et al., 2024).

Research Question:

Using supervised statistical learning, what is a location that, in year 2025, reflects the flood risk of a selected North American city in the year 2050?

Section II: Methodology

2.1 Definition of Flood Risk:

With respect to the research question stated in Section I, this study defines flood risk as follows:

- The probability of getting a flood (flood = 1) in a location.
- The probability of getting a Category X flood (Category = 1, 2, 3) in a location, conditional on the fact that a flood has already happened.
- The number of people that will die in a location, conditional on the fact that a flood has already happened.
- The number of people that will be displaced in a location, conditional on the fact that a flood has already happened.

Given the four above definitions of flood risks, there will be four predictions:

1. What is the probability of getting a flood in each grid of the 2025 raster and the 2050 raster?
2. What is the probability of getting a Category X flood in each grid of the 2025 raster and the 2050 raster, conditional on the fact that a flood has already happened in each grid?
3. What is the number of people that will die in each grid of the 2050 raster, conditional on the fact that a flood has already happened in each grid?
4. What is the number of people that will be displaced in each grid of the 2050 raster, conditional on the fact that a flood has already happened in each grid?

2.2 Geospatial Data Organization

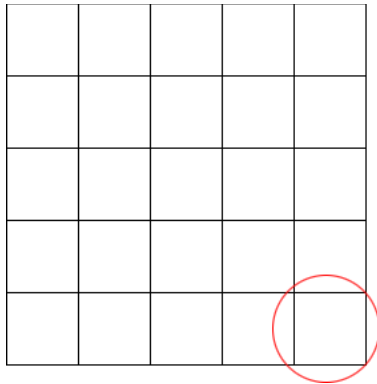


Figure 1: A basic raster for one time period. The circled grid signifies one value at one specific time.

This study will utilize the concept of rasters. Used in geospatial analyses, a raster defines space as grids of equal size, with each grid bearing a single value. Illustrated in Figure 1, one raster layer represents one value at one specific time. To depict the change of values over time, there needs to be multiple raster layers. In this study, one raster layer represents one month.

In R, rasters are often converted from simple feature (sf) objects, which are dataframes with an added column for geometry. With different coordinate reference systems, the geometrical coordinates can bear different values. The most standardized and common geometrical coordinates for geospatial modelling are latitudes and longitudes, which, in this study, is referred to as EPSG:4326.

2.3 Building 2025 Rasters:

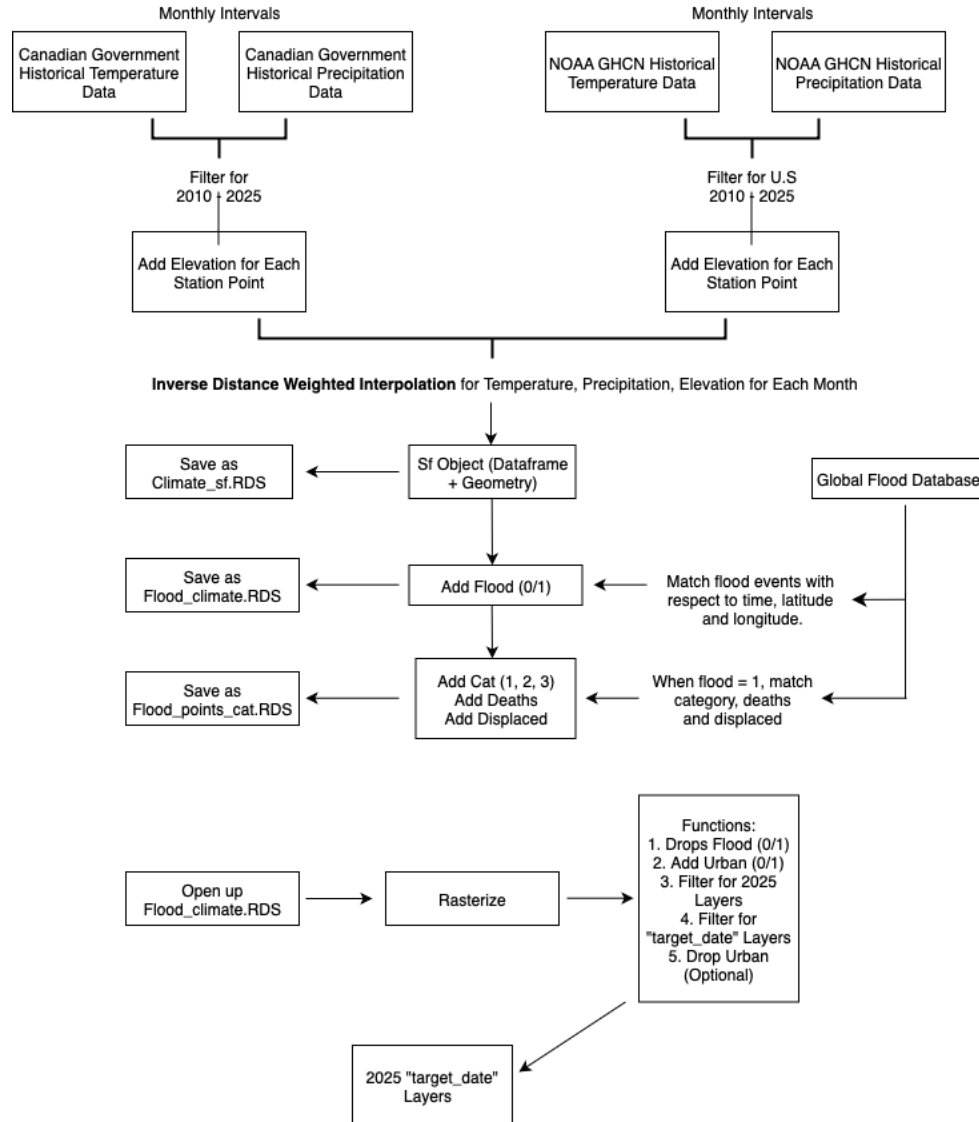


Figure 2: Methodology for building 2025 rasters.

With reference to Figure 1, the goal of data wrangling in this section is to transform observational data of Canada and the U.S. into a layer of rasters for a specified month in 2025. These layers represent the variables of temperature, precipitation, elevation and the optional variable of urban. The structure of one set of monthly raster layers is illustrated in Figure 2 as an example.

Sourced from Environment and Climate Change Canada (ECCC), and National Oceanic and Atmospheric Administration (NOAA), historical observed temperature and precipitation data were obtained for all monitoring stations within Canada and the United States¹ for the years 2010 to 2025. While Canadian data is more accessible in the CSV format, NOAA data is available only in fixed-width file format, which requires the user to input the correct number of characters for each column to unlock. Data for the Alaska region is scarce, and observations from NOAA do not extend to beyond July 2010, resulting in the entire state being excluded in later analyses.

¹ For this study, the author has defined United States boundaries to be within North America, therefore excluding temperature, precipitation and elevation observations for the Virgin Islands and Hawaii.

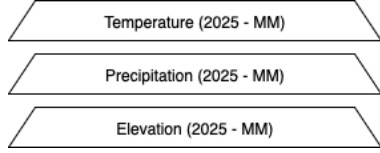


Figure 3: The structure of 2025 monthly raster layers.

As temperature and precipitation observations are recorded by monitoring stations, there are less than 10,000 points scattered on a map of North America, excluding Alaska at each month. To create the raster layers of a single month based on these coordinate points, temperature, precipitation, along with the corresponding elevation points of each monitoring station, must be interpolated to cover all areas missing from observational data coverage. Since station points are not evenly distributed, most coverage concentrates in the southern United States, and only one

station is responsible for the tip of the Queen Elizabeth Islands. The most suitable interpolation method – thin plate spline – used for irregular distributions of data was incompatible with a MacBook Air laptop’s processing power, which led to the adoption of the inverse distance weighted (IDW) interpolation method instead. Though the interpolation was implemented smoothly for all variables, the following two conditions of IDW interpolation were not met in this study:

1. The average distance cannot be greater than the highest input or less than the lowest input, therefore extremes must also be sampled in your observational data.
2. The sampling distribution must be sufficiently dense with regards to local variation surrounding the interpolated point.

For further identification of errors, refer to Section IV Discussions. Given observations (s_i, z_i) of temperature, precipitation or elevation value z_i at location s_i and prediction location s_0 , the estimated value $z(s_0)$ is:

$$z(s_0) = \sum_{i=1}^n w_i(s_0) z_i \quad (2.1)$$

$$w_i(s_0) = \frac{d(s_0, s_i)^{-p}}{\sum_{j=1}^n d(s_0, s_j)^{-p}} \quad (2.2)$$

Thus, all interpolated values of North America are obtained via Equations 2.1 and 2.2, in which the estimated value of a predicted location is based on the respective weights (w_i) of many predicted values at that location. Weights of each prediction is determined by the inverse power of the distance from the observed location to the predicted location, compared to distances from all other observed locations. Consequentially, the name of the IDW method emphasizes inverse power due to the logic that closer points have larger weighting than points with a longer distance. However, without the fulfillment of the above two conditions, the weighting assigned to sparse, uneven distributions of observations in Equation 2.2 will result in a heavily biased estimated value at the predicted location.

Furthermore, there is no database currently available that tracks all observed flood events in both Canada and the United States from 2010 to the present time. As of August 2025, the University of Colorado’s Flood Observatory (DFO) is the most comprehensive compilation for the last four decades – it is global in scale and identifies flood observations, along with subsequent flood severity categories, deaths and displaced data from news reports. Its only deficiency is the tracking of solely large flood events, which the DFO defines as floods that have incurred significant damage on agriculture and infrastructure. Smaller-scale flash floods or pluvial floods² are not recorded in the DFO archives. Though the methodology is unclear, flood events are categorized into severity indices of 1, 1.5 and 2, which increases in scale due to the damage incurred from each event.

² Flash floods are the result of heavy, torrential rain in a short period while pluvial floods occur due to the inability of a city’s drainage infrastructure to absorb excessive surface water.

Given the variables of flood category, death and displacement for each flood observation, raster layers are then built for model training and model validation, as referenced in Figure 2 and Figure 5.

In order to facilitate the rasterization process, all previous interpolated temperature, precipitation, elevation, along with the DFO's flood archive are converted into sf objects, where all variables above are matched according to their geometries. For the training process of the flood prediction model, flood is categorized as having occurred or not occurred for each geometrical point, along with its corresponding temperature, precipitation, elevation, at each monthly time step from 2010 to 2025. For each month, there are separate layers for temperature, precipitation, elevation, flood occurrence, severity category, death, and displacement. However, due to the computational limitations of a MacBook Air, the resolution for each raster layer remains low at 0.22° latitude and longitude. Since there are many geometrical points within a grid of 0.22°, the final value of the grid for each variable layer will be calculated as follows:

- Grid temperature, precipitation and elevation are calculated as an average of their respective values at each geometrical point.
- Flood occurrence becomes 1, if there happens to be a value of 1 at any geometrical point within the grid.
- Flood severity category takes the highest categorical value that occurs within the grid.
- Death and displacement are calculated as the sum of the values attached to the geometrical points within the grid.

The purpose of the 2025 raster layers is to predict the four flood risks, as referenced in Section 2.1, of a particular month once the supervised machine learning models are trained. From the above sf object, functions can filter for the month of interest by 1) dropping the flood occurrence variable, 2) adding an indicator variable of urban or rural area to control for bias in death or displacement prediction, 3) filtering for 2025, 4) filtering for the monthly layers of interest, and 5) dropping urban if the variable is not needed.

2.4: Building 2050 Rasters:

The data source for the 2050 raster layers is the North American Coordinated Regional Downscaling Experiment (NA-CORDEX) regional climate model outputs. As the 2025 layers are built with coordinate reference system EPSG:4326, which maps geometries according to corresponding latitude and longitude degrees, the outputs selected from NA-CORDEX also must be within the same EPSG:4326 format, because reprojection of geometries to other coordinate reference systems is out of the computing range of a MacBook Air. Currently, only outputs of the following models have the desired coordinate reference system:

1. Geophysical Fluid Dynamics Laboratory – Earth System Models Version 2M (GFDL-ESM2M)
2. National Center for Atmospheric Research – Weather Research and Forecasting Model (WRF)

Though the only available representative concentration pathways (RCP) scenarios are 4.5 and 8.5, this limitation on model output is befitting of potential RCP scenarios available due to currently global climate policy. Thus, there are two versions of 2050 monthly raster layers that reflect the concentration of greenhouse gases (GHGs) in the atmosphere:

1. RCP 4.5: Business-as-usual global climate policy will result in a radiative forcing³ of 4.5W/m² by 2100.
2. RCP 8.5: No global climate policy for alleviating GHG accumulation in the atmosphere, resulting in a radiative forcing of 8.5W/m² by 2100.

After selecting GFDL-ESM2M model outputs for both RCP scenarios, the downloaded NetCDF files are then converted to raster layers of temperature, precipitation and elevation. Following procedures similar to

³ RCP radiative forcing is expressed in units of Watts per square meter.

Section 2.3, functions can filter for the future month of interest by 1) attaching a date to every monthly layer, 2) filtering for the year 2050, 3) filtering for the specific month of interest, and 4) adding urban as a variable if needed. The raster layers should also have a resolution of 0.22° latitude and longitude in order to match with the 2025 raster layers, as the same machine learning models will be used to predict for flood risk in both time periods.

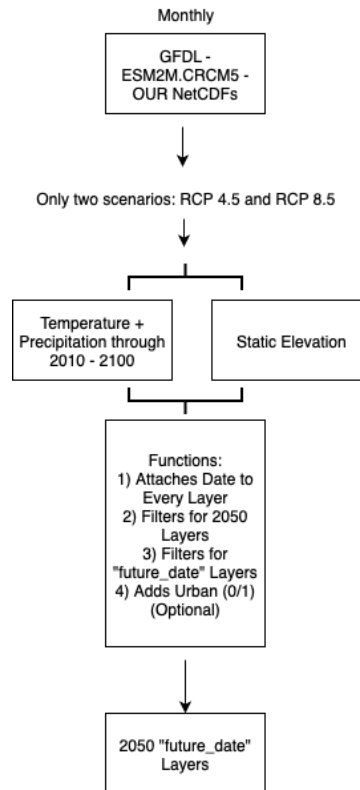


Figure 4: Building 2050 rasters.

2.5: Model Training

Supervised machine learning models differ from unsupervised models in that they need training from labelled output and input data predict outputs in new, unseen data. Referring back to the four flood risk definitions in Section 2.1, each definition would be an output with inputs of temperature, precipitation, elevation, as well as the optional predictor of urban.

Figure 5 shows the process of building training data with both raster layers and sf objects. Depending on one's computing limitations, training models with raster layers is less computationally intensive than with sf objects, as there are 700,000 grids compared to 3,000,000 geometric coordinate points.

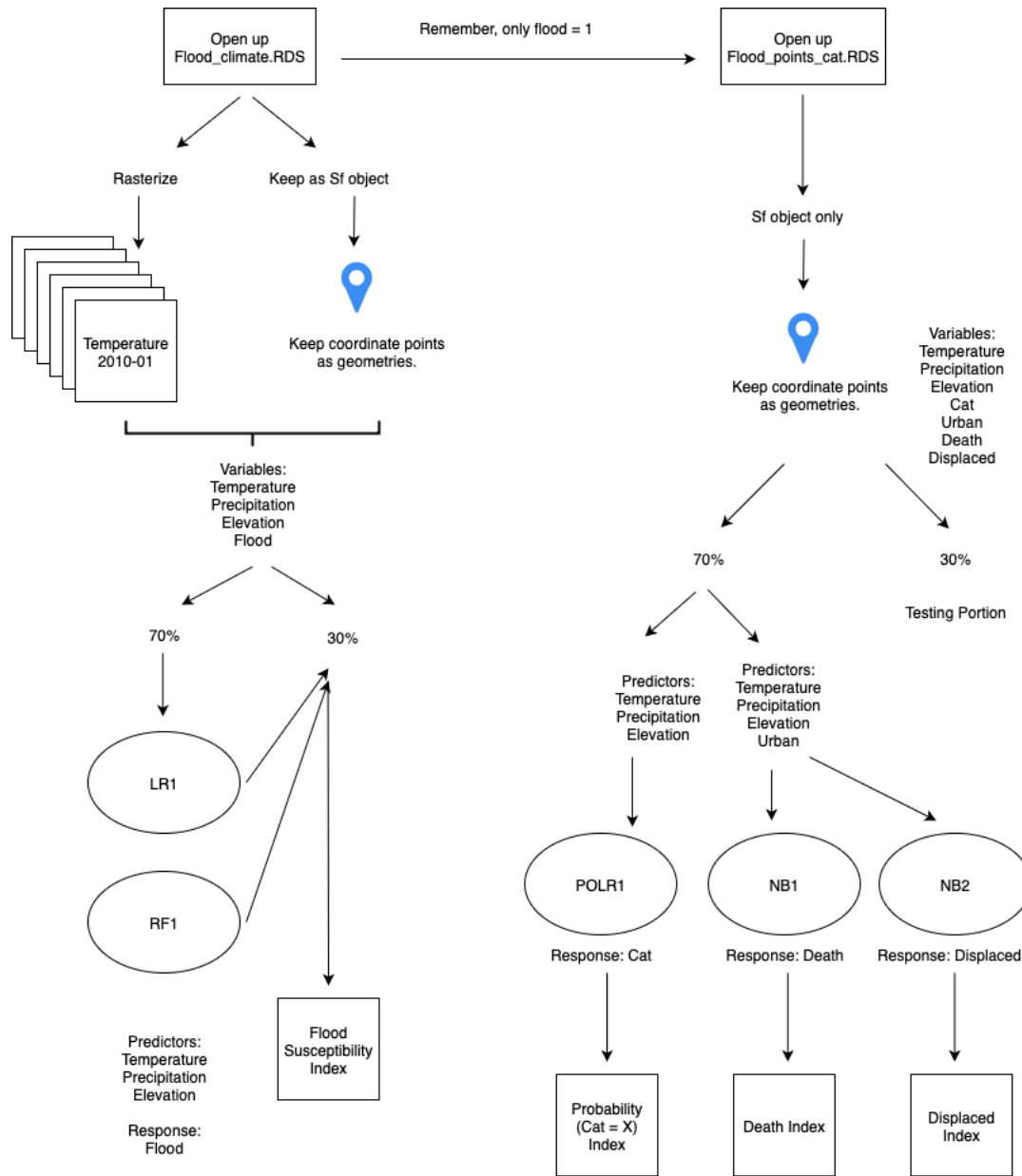


Figure 5: Process for regression model training.

Due to the limitations of the outputs available from NA-CORDEX's climate model outputs, the training of all supervised machine learning models is constrained to the usage of the following inputs: temperature, precipitation, elevation and urban. Refer to Section 4 for more discussions on prevalent errors in model training and potential improvements in future studies.

Logistic Regression (Binary Outcomes)

Referring to Figure 5, model training can be performed with both sf objects and their rasterized layers. Using the "flood_climate" sf object and all the historical 2010 – 2025 raster layers, a logistic regression model with binary outcomes can be created to predict for the probability of future flood occurrence, $P(\text{flood} = 1)$. Labelling flood occurrence as the output variable, in which flood = 1 for a flood having occurred, and flood =

0 for no flood event, a logistic regression with predictors of temperature, precipitation and elevation. Historical data was split as 70% for training and 30% for testing.

$$\log\left(\frac{P(\text{flood})}{1 - P(\text{flood})}\right) = \beta_0 + \beta_1(\text{temperature}) + \beta_2(\text{precipitation}) + \beta_3(\text{elevation}) \quad (2.3)$$

Random Forest (Binary Outcomes)

Similar to the logistic regression above, a random forest made up of 100 decision trees can also determine the probability of a flood occurrence. While Figure 6 demonstrates a general random forest with binary options at each split, the decision trees used in model training splits at random values of temperature, precipitation and elevation. A class outcome of a flood event (flood = 1) or no flood occurrence (flood = 0) is determined at the leaf node of every decision tree, and the majority outcome – also known as bagging – of 100 decision trees is then defined as the final prediction of class. As this study is looking for the probability of a flood occurring, class prediction can be used to calculate flood probabilities from each decision tree in Figure 7.

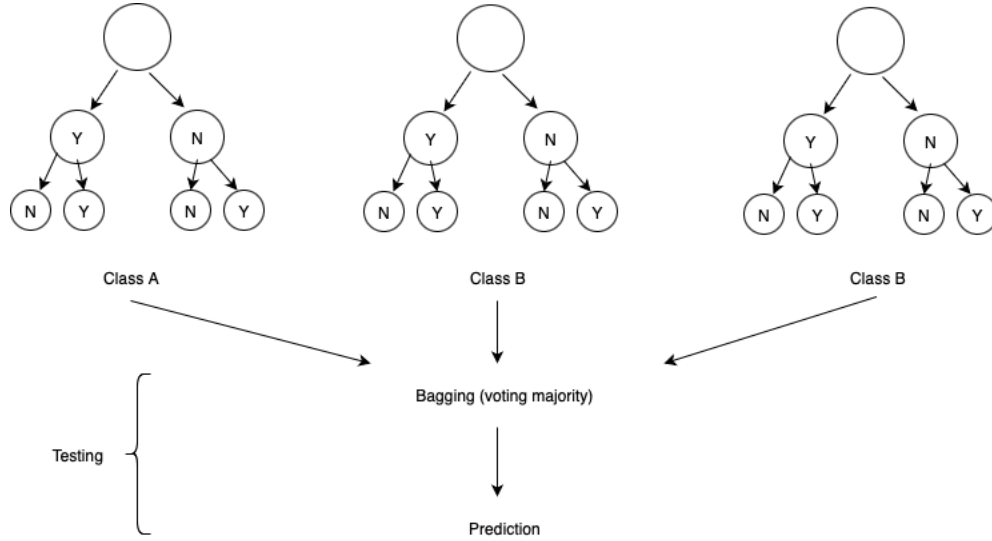


Figure 6: A general random forest, made up of 3 decision trees. Class is the result in this example.

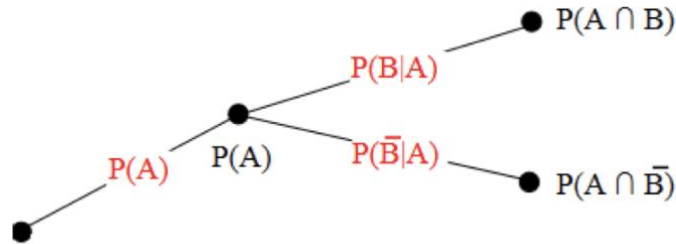


Figure 7: Probability within a decision tree.

Comparing both the logistic regression and the random forest, the model with the highest predictive performance should be used for the prediction of flood events.

The Area Under the Receiver Operating Characteristic Curve (AUROC) plots the true positive rate against the false positive rate. Using testing data to assess predictive performance, a high AUROC score close to 1.0 and above 0.5 signals that the model can separate true positive from the false positive at various classification thresholds. However, an AUROC assessment should always be coupled with an assessment of overfitting as well, as a perfect predictive performance of 1.0 can be a signal of the model fitting too well to its training data.

Ordinal Logistic Regression

The following regression analyses will use the sf object “flood_points_cat”, as indicated in Figure 5, where only geometrical points with a flood event are included. Since floods have already been categorized by the DFO into 3 levels of increasing severity, an ordinal logistic regression model will be trained with flood categories as the output. As the choice of input variables is limited, this regression will use the same predictors of temperature, precipitation and elevation as the previous logistic regression.

$$\text{logit}(P(Y \leq j|X)) = \alpha_j - (\beta_1(\text{temperature}) + \beta_2(\text{precipitation}) + \beta_3(\text{elevation})) \quad (2.4)$$

Poisson Regression

Both death and displacement data are presented as count data, also known as positive integers. In this case, a linear regression is not recommended to train for the prediction of future deaths and displacement counts, due to the possibility of decimals in the outputs. However, the choice of Poisson regression is not the most fitting for death and displacement data, because the expected mean must be equal to the variance for each distribution of death or displacement at each respective geometric point. The best choice would be a negative binomial regression, which usually models data with overdispersion, but this model could not be processed in R Studio with errors present in the design of the sf object.

$$\begin{aligned} \log(E[\text{displaced}]) \\ = \beta_0 + \beta_1(\text{temperature}) + \beta_2(\text{precipitation}) + \beta_3(\text{elevation}) \\ + \beta_4(\text{urban}) \end{aligned} \quad (2.5)$$

$$\begin{aligned} \log(E[\text{death}]) = \beta_0 + \beta_1(\text{temperature}) + \beta_2(\text{precipitation}) + \beta_3(\text{elevation}) \\ + \beta_4(\text{urban}) \end{aligned} \quad (2.6)$$

2.6: Predictions of Flood Risks

Figure 8, 9, 10 all illustrate prediction processes for the following four flood risks for both the 2025 and 2050 raster layers:

1. Probability of a flood event occurring using logistic regression or random forest trained with binary output.
2. Probability of a category X flood, given that a flood has already occurred, using ordinal logistic regression named “POLR1”.
3. Number of deaths, given that a flood has already occurred, using Poisson regression named “NB1”.
4. Number of displaced people, given that a flood has already occurred, using Poisson regression named “NB1”.

For prediction 1, the model with the highest AUROC score should be selected if there's no concern for overfitting. The remaining models should be critiqued based on their respective residual deviance, but further improvements are limited by the number of predictors available to both the 2025 and 2050 raster layers.

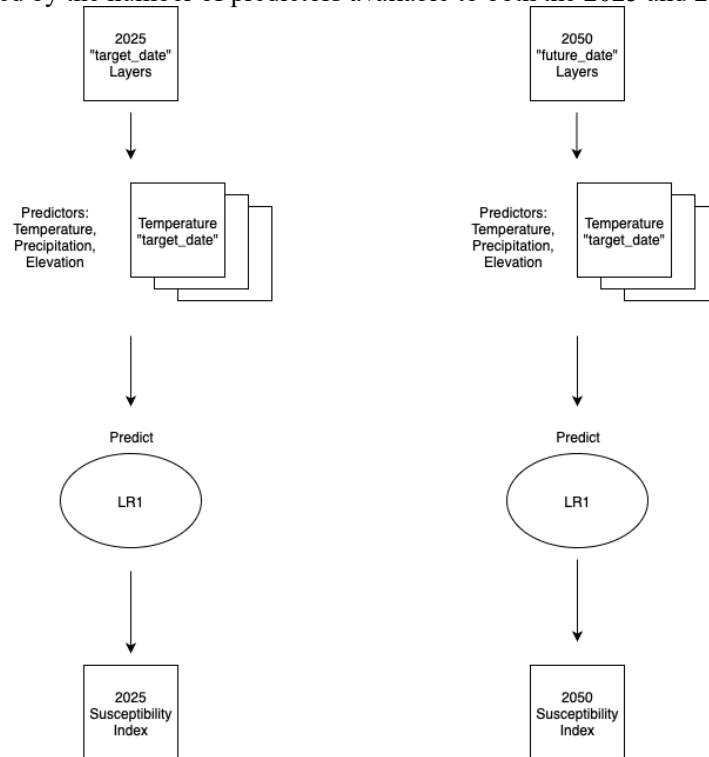


Figure 8: Prediction of the probability of flood occurrence for both 2025 and 2050 rasters

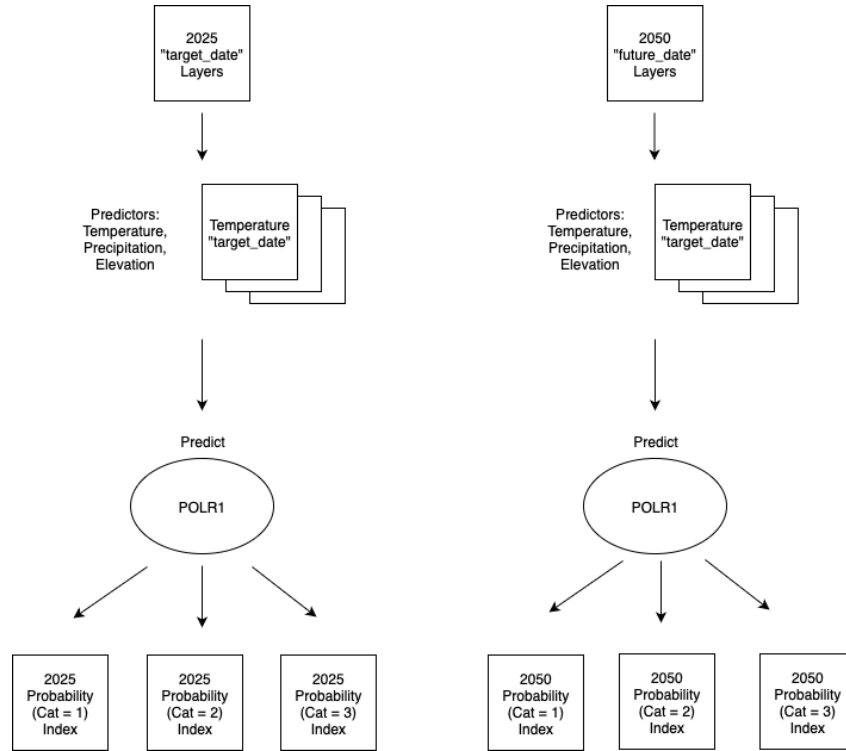


Figure 9: Prediction of the probability of getting a category X flood, given that there is a flood, for both 2025 and 2050 rasters.

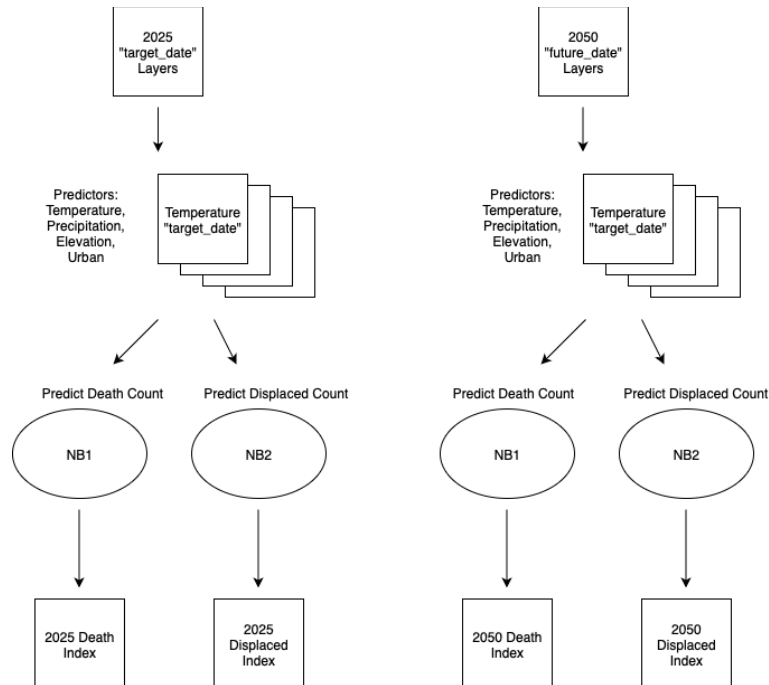


Figure 10: Prediction of death and displacement counts, given that there is a flood, for 2025 and 2050 rasters.

2.7: Analogue Calculation

As elaborated above in Section I, analogues are defined as locations in 2025 that reflect the flood risk of a target city in the year 2050. Referencing Figure 11, a grid containing the geometric boundaries of the

target city in the 2050 susceptibility raster will convert its flood risk probability into a scalar value. Taking the absolute difference between the 2025 susceptibility raster and the scalar value, a raster of absolute differences from the target city's flood risk in 2050 will be calculated. As a result, the analogue location is the grid with the smallest absolute difference value.

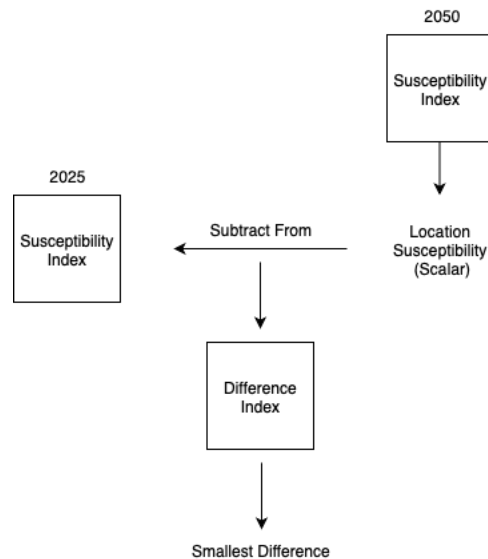


Figure 11: The process of calculating analogue locations.

Based on all the processes illustrated above, the following model assumptions can be defined for the predictive models.

2.8: Model Assumptions:

Based on the research question, an analogue location would be a particular grid on the raster that bears the same flood risk, with respect to the definitions listed in Section 2.1. Thus, flood risk is considered as a probability rather than a predicted result of a flood not occurring (Flood = 0) or occurring (Flood = 1).

Assumptions in model training and prediction of future risks are as follows:

- Selection of probability rather than “class” when running predictions.
- Assumption of only two possible greenhouse gas concentration scenarios for policymakers at the current time: RCP 4.5 and RCP 8.5.
- Elevation, temperature and precipitation 2025 rasters are derived from interpolation instead of actual observations.
- The timeline of 2010 – 2025 is assumed to reflect true long-term trends in climate change.
- The predictors of temperature, precipitation and elevation is assumed to have the same effects on the response variables of flood, cat, death or displaced in 2050 as compared to 2025.

Section III: Results

3.1: Model Validation and Predictive Performance

Predictor Effect Estimates	Logistic Regression 1	Logistic Regression 2	Ordinal Logistic Regression	Poisson Regression (Death Count)	Poisson Regression (Displacement Count)
Temperature	0.03345	0.034081	0.030951	-0.0166883	0.08469
Precipitation	-0.05789	-0.052562	-0.033508	-0.0246164	-0.003872
Elevation	-0.001536	N/A	-0.001186	-0.0113683	-0.003476
Urban	N/A	N/A	N/A	-2.1143822	-3.503
AIC	97197	99293	11694.79	10067	13180680
Residual deviance	97189 on 2141042 degrees of freedom	99287 on 2141043 degrees of freedom	11684.79	9353.1 on 7782 degrees of freedom	13160262 on 7782 degrees of freedom

Table 1: Effects of temperature, precipitation, elevation and urban as predictors, as well as Akaike Information criterion. All predictors showed significance of $p < 0.05$.

After training logistic regressions 1 and 2 with binary responses, all predictors of temperature, precipitation and elevation had significant effects on the logit response. The removal of elevation as a predictor in regression 2 resulted in little change in the effects of temperature and precipitation, signaling that elevation is not correlated with other predictors. Though multicollinearity is not a concern with these three predictors, the little reduction in AIC with the addition of elevation as a predictor shows that there is a lot of unaddressed variation in the training data. More predictors should be added to reduce residual deviance.

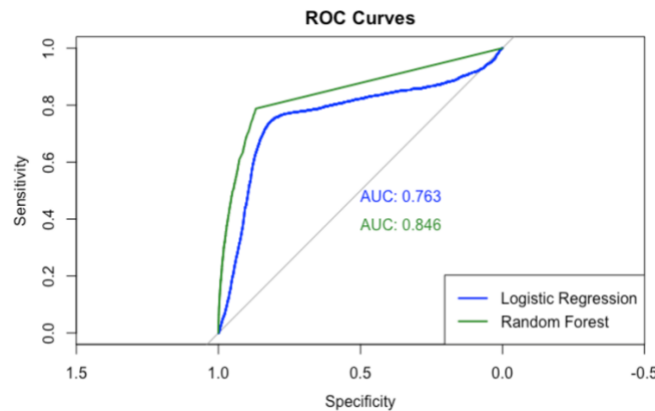


Figure 12: AUROC score comparison between logistic regression and a random forest of 100 trees

With reference to Figure 12, the random forest of 100 decision trees performed better than logistic regression 1 in terms of their respective AUROC scores, signaling that the random forest can discriminate between true positive predictions and false positive predictions slightly better than the logistic regression. However, the AUROC score of both is still under 0.9, which could be increased with more predictors added to both models. After selecting the random forest for the prediction of flood occurrence probability, a test for overfitting was conducted by identifying the accuracy of the model identifying true positives in both testing and training data.

Threshold	Training Accuracy	Testing Accuracy
$P(\text{flood} = 1) = 0.5$	0.442065	0.432789
$P(\text{flood} = 1) = 0.1$	0.998255	0.990315

Table 2: Overfitting test for the random forest of 100 trees.

A threshold transforms the probability response into a class output, in that a probability over the threshold would be defined as a flood event. If overfitting exists, the model should be more accurate in

identifying true positives in the data from which it was trained as compared to its accuracy in the testing data prediction. For both thresholds in Table 2, training accuracy is roughly equal to the testing accuracy, resulting in no evidence of overfitting.

A concern in the ordinal logistic regression is the p-value of 0 for precipitation and elevation. Though the interpretation of such a low p-value can indicate a significant effect of the two predictors, it is likely the consequence of complete separation of the categorical outcomes, or a violation of the proportional odds assumption. In the case of complete separation, predictors can perfectly distinguish between outcome categories. As there are only 3 categories of flood severity, a specific value for either precipitation or elevation can directly indicate a higher severity category, resulting in a p-value of 0 in the estimation of the effects of each predictor on that category. At the same time, the proportional odds assumption posits that the relationships between each predictor and the odds of being in a higher flood severity category is the same across all categories. If this assumption is violated, then the estimated effect of each predictor for each category is different and the p-value would be misleading in its indication of effect significance.

Comparing death and displacement data, Table 1 shows an abnormally large residual deviance value in the Poisson regression fit for displacement data compared to the fit for death data. This difference indicates that there is a lot of unaddressed variation in the displacement data that could not be explained by the predictors of temperature, precipitation, elevation and urban. The factors that result in displacement should be explored further.

3.2: Analogue Index

City	Analogue Location for Flood Probability	Analogue Location for a Category 3 Flood	Death Count in 2050	Displacement Count in 2050
Miami, Florida, USA	Houston, Texas	Annapolis, Maryland	0	200
Brockville, Ontario, Canada	Somewhere on Ellesmere Island	West Bay, Labrador, Canada	0	0
Vancouver, British Columbia, Canada	Bridgeport, Connecticut, USA	Carson City, Nevada, USA	0	100
Sydney, Cape Breton Island, Nova Scotia, Canada	Somewhere on Ellesmere Island	Eureka, California, USA	2	100
Crescent City, California, USA	Jacksonville, Florida, USA	Haida Gwaii, British Columbia, Canada	0	0

Table 3: Index of analogues for five distinct locations in North America.

Table 3 showcases flood risk analogues for the month of June under the scenario of RCP 8.5. Due to the small sample size of flood events from 2010 to 2025 available in the DFO archives, the supervised machine learning models are badly trained and badly fit, resulting in some analogue locations indicating unrealistic flood risks. For example, two target cities above indicate that their respective analogue locations are on Ellesmere Island, where flooding rarely happens. The only plausible explanation is that the trained regression models predicted a nearly 0 probability of flood occurrence in the 2050 raster layers, and thus the grid with the smallest absolute difference in 2025 is Ellesmere Island, where probability of flood occurrence is also nearly 0.

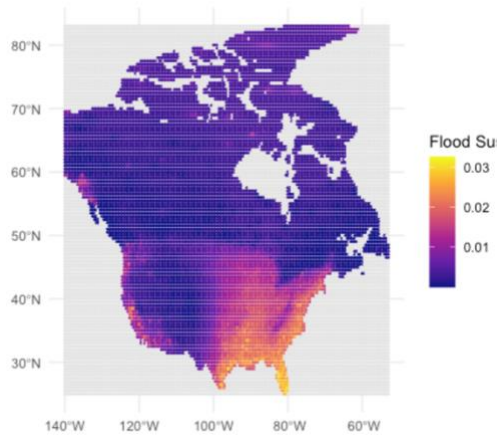


Figure 13: Probability of flood occurrence in June 2025.

With reference to Figure 13, the probability of flood occurrence is shown to be nearly zero nearly the entirety of Canada. The distinct border between Canada and the U.S. confirms that the inverse distance weighted method for interpolation is not the best choice for this study. As Canada and U.S. data are sourced from two governmental databases, temperature, precipitation and elevation values at the border had slight differences at the time of download. Canada also had a scarce amount of monitoring stations compared to the U.S., with most stations installed at the border. An interpolation that depended on evenly distributed observations resulted in Canada having less variation than the U.S. throughout the country. The slight differences in temperature, precipitation and elevation became more exaggerated with this interpolation method. The next section explores methods that could improve these prediction results.

Section IV: Discussion and Conclusion

4.1: Applications of Flood Risk Analogues

Though flood occurrence is still uncertain, supervised machine learning models can provide a preliminary analysis of a location's risk profile by assessing its future risk based on past flood events. Flood risk analogues further improve the understanding of risks by providing an example of an analogue location that is dealing with similar risks at the present time.

For policymakers, flood risk analogues not only help local and federal government to prioritize regions at risk in the next decade, but also provide a way to communicate climate risks to the public in simple terms. By exhibiting a realistic example of flood risk, governments can conduct scenario analyses for climate policies, such as no GHG control, business-as-usual or ambitious GHG control, to examine their respective effects on risk mitigation in the near future.

For urban planners, flood risk analogues can aid in the design of resilient cities. Given a reference point for future disasters, cities can adjust infrastructure codes, such as in storm drains, levees and building elevations, in preparation of future risks. Planners can also guide the drafting of new zoning policies to ensure the sustainability of new real estate development proposals. Similarly, planners can also assist in risk mitigation by incorporating nature-based solutions (NbS) in city design. For example, if soil moisture was a predictor shown to have a significant negative effect estimate on the response of flooding probability, then planners could design conservation areas for the preservation of soil health in order to reduce future flood risk.

For home insurers in the property and casualty field, the analogue location's risk profile can serve as a reference for ratemaking. Past loss ratios from the analogue location can inform loss projections for the target city, and this method is discussed further in Section 4.2. Furthermore, marketing teams can explore new campaign strategies to guide policy sales based on the development of new zoning policies.

4.2: Potential Improvements

A flood risk analogue's reliability is based entirely on the supervised machine learning models used for the prediction of the probability of flood occurrence, death count and displacement count. Due to limitations in computing power, variable selection, flood data availability, the models trained for prediction had many flaws, which led to unrealistic analogue locations in Section 3.2. Past studies of flood risk using random forests for prediction, such as Wahba et al. (2024), include predictors such as snow amount, soil moisture, as well as an indicator variable for the presence of nearby watersheds. However, the selection of predictors is always limited by the available variables in climate model outputs.

NA-CORDEX was the only database that offered a comprehensive list of North American regional models, as of August 2025. The highest resolution offered is 0.11° latitude and longitude, but this option can only be explored with better computing power. With more research and development efforts in climate modelling, there may be more output variables available for modelling flood risk.

Furthermore, the addition of new predictors would impact multicollinearity. While it was shown in Section 3.1 that no significant multicollinearity was present between temperature, precipitation and elevation, the addition of snow amount would correlate with the precipitation predictor for northern regions and the winter season, resulting in a high variance inflation factor. However, this problem can be ignored if the variable for snow amount can be transformed into a power or product of precipitation, as the p-value for estimated effect of this predictor on the response is not affected by multicollinearity. (Allison, 2012)

At the same time, the problem of complete separation may also be present for both the binary logistic regression and ordinal logistic regression with the addition of the indicator variable of nearby watershed

presence. Past research has shown that locations with watersheds are more susceptible to flood events (Ozcelik and Benli, 2025) and may perfectly predict the occurrence of flood or the severity category of flood events.

With better computing power, thin plate splines should have been used for interpolation instead of the inverse distance weighted method. As a multivariate version of smoothing splines, thin plate splines aim to fit a smooth surface through unevenly distributed data points. Figure 14 demonstrates the analogy of thin plate splines bending a thin sheet of metal through data points with minimal bending energy.

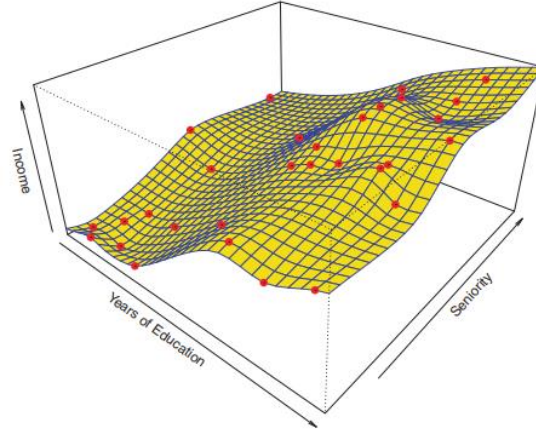


Figure 14: Thin plate spline interpolation illustrated. Source: StackOverflow

Equation 4.1 demonstrates the process for the prediction of temperature, precipitation or elevation at a new location $f(x, y)$. The predicted value is a combination of weighted radial adjustments centered at each station point, $w_i r_i^2 \log r_i$, based on a plane of $\alpha_0 + \alpha_1 x + \alpha_2 y$.

Weights, w_i , are found by solving a linear system that balances the curves centered at each data point with the above plane. The final product is a fitted surface that passes smoothly through all the station points with a smoothness parameter of λ .

$$f(x, y) = \alpha_0 + \alpha_x x + \alpha_y y + \sum_{i=1}^N w_i r_i^2 \log r_i \quad (4.1)$$

As the DFO of the University of Colorado used news articles to gather large flood event observations, and the associated severity, death and displacement counts of each flood recorded, future studies of flood risk analogues should also use a similar method to gather a more comprehensive list of flood observations that would result in a larger sample size of data to be used for model training. Using a simple web scraper on major Canadian and American news sites, such as the Canadian Broadcasting Corporation and the New York Times, coordinates, duration, severity, physical damage, death and displacement counts of all flood events that occurred after January 1st, 2010 can be collected. If the physical boundaries of the flood event are unclear, the buffer function in the sf package can create a circle that reflects the area of flood damage. In this case, a more comprehensive list of flood events that includes smaller-scale flash floods or pluvial floods, along with the addition of new predictors, would result in logistic regression models that have less residual deviance.

Another innovative way to predict flood risk would be to abandon the concept of parametric relationships, which includes linear regressions and logistic regressions. Floods in the real world are rare, extreme events, so applying a “global”, fixed coefficient to an entire distribution of temperature is irrational, as historical flood events likely share a nonlinear relationship with temperature. Furthermore, floods are location-dependent, so a “local” approach to prediction is another avenue to explore. Incorporating similarity-based models instead of linear models, these models can predict future flood risks by looking at similar past cases.

An example of a similarity-based approach would be the case-based reasoning method, which has been applied in previous studies of flash flood risk (Fernandez et al., 2022). If Brockville, Ontario, Canada is set as the target city, then its flood risk for June of 2050 would be reflected in locations with similar temperature, precipitation and elevation values in the June monthly averages of the past 15 years. In Equations 4.2, X_i represents a vector of conditions for the analogue location in June of various years, while X^* represents a vector of Brockville's conditions in June 2050, with elevation remaining constant over time.

$$\begin{aligned} X_i &= (temp_i, precipitation_i, elevation_i) \\ X^* &= (temp_{2050}, precipitation_{2050}, elevation_{Brockville}) \end{aligned} \quad (4.2)$$

Once a list of these analogue locations is found, their past flood occurrences would be recorded as a binary variable of yes/no. As indicated in Equation 4.3, the Mahalanobis distance between Brockville's future conditions and the analogue location's past conditions can be found. Similar to the Euclidean distance, the Mahalanobis distance adjusts for correlation between variables, as well for large variations in differences. It prevents large effects on distance from specific variables. For example, a distance of 10 may be an insignificant difference to precipitation, which is measured in terms of millimeters, while a distance of 10 in degrees Celsius has significant effect.

$$d_{mah}(X^*, X_i) = \sqrt{(X^* - X_i)^T \Sigma^{-1} (X^* - X_i)} \quad (4.3)$$

The Mahalanobis distance allows for the incorporation of the K nearest analogue method, where a number of analogue locations with the closest conditions to Brockville in June 2050 are picked. Weights are then assigned to flood risks. In Equations 4.4 and 4.5, risks from closer locations influence Brockville's future risk more than risks from further locations. The parameter σ can be customized to reflect the decay of weights with distance. With N analogue locations that each contain the binary flood occurrence response of y_i , a probability, p, for Brockville's future flood occurrence can be found by calculating the fraction of locations with flood occurrences out of all locations.

$$w_i = e^{-\frac{d_i^2}{2\sigma^2}} \quad (4.4)$$

$$p = \frac{\sum_N w_i y_i}{\sum_N w_i} \quad (4.5)$$

For property and casualty insurers, flood risk analogues can provide a reference for ratemaking, as mentioned in Section 4.1. In the loss development method, loss development factors (LDFs) are used to project the ultimate, final cost of a claim. In the absence of historical data for flood events in a target city, an analogue location can provide a plausible empirical benchmark for future claims, given similar market coverage in both cities. After adjusting for prospective inflation, exposure unit, the historical loss development factors of the analogue city can be used in credibility weighting or scenario analysis, where analogue LDFs can be an extreme scenario of no GHG mitigation. In Equation 4.6 where a credibility weighting is conducted, weight is determined by actuarial judgement.

$$LDF_{blended} = w * LDF_{target} + (1 - w) * LDF_{analogue} \quad (4.6)$$

Overall, flood risk analogues can provide many benefits to the public and the private sector in its applications to policymaking, urban planning and insurance. However, analogues of flood risks are based on predictions from linear models, which are inherently flawed in their fit to rare, location-dependent events like floods. Though there are currently limitations on data availability and climate model outputs, there are still many potential ways to improve and build upon this preliminary exploration into North American flood risk analogues.

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